United International University

Department of Computer Science and Engineering

Final Exam, Summer 2025

CSE 4531: COMPUTER SECURITY

Total Marks: 40, Duration: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Answer all the questions.

1.	a) Bob and Simon are working on a secure messaging app for their university project. They want to make sure that whenever Bob sends a message to Simon, he can verify that the message truly came from Bob and that it was not modified during transmission. To achieve this, they decide to use a special **Authentication** mechanism. Explain how this particular mechanism helps Bob and Simon ensure both message integrity and authenticity in their communication. Support your explanation with a suitable **diagram** showing the working process of the mechanism.	**3**
	b) What is a Ransomware attack? What is its impact on the victim? How can you prevent such an attack?	**3**
	c) What is the number of 10-character passwords that could be constructed from a character set that includes all 128 characters in the ASCII character set. How long, on average, would it take an attacker to guess such a password if he could test a password every microsecond?	**4**
2.	a) Consider that you have just got a call from the University Hall administrator, who notified that either you or your roommate is issuing denial-of-service attacks against other students from your shared network segment. You know you are not doing this, but you are unsure about your roommate. How can you tell if this accusation is true or not? And if it is true, what should you do about it?	**3**
	b) Explain stack smashing attacks with an example. How to protect such attacks?	**3**
	c) How is an encrypted virus formed by the attackers? How can you detect encrypted viruses? Explain briefly.	**4**
3.	a) Explain CSRF attack on a website with necessary steps. What is its impact on the victim?	**3**

	b) A website displays user-submitted content (e.g., comments, reviews, or profile bios). An attacker posts a comment that contains malicious JavaScript. Whenever any user views the page, the malicious script runs inside their browser. Identify the **type of attack** described in this scenario and explain how it works. List **three** defensive controls the web team should apply (briefly justify each).	**3**
	c) How can an attacker launch a TCP SYN flooding attack? Explain with respect to the message flow diagram. How can you mitigate such an attack?	**4**
4.	a) Calculate the following big moduli: $101^{735} \, mod \, 17$	**3**
	b) What is the main role of Certification Authority (CA) in digital signature? What will happen if CA is absent?	**3**
	c) Alice wants to use **RSA** for secure communication. She chooses two prime numbers, p=17 and q=13. Determine a suitable public key (e) for Alice (Take the smallest possible key permitted by the algorithm), the private key (d). Then, encrypt the message m=43 with the public key to compute the ciphertext. And verify that decryption retrieves the original message.	**4**